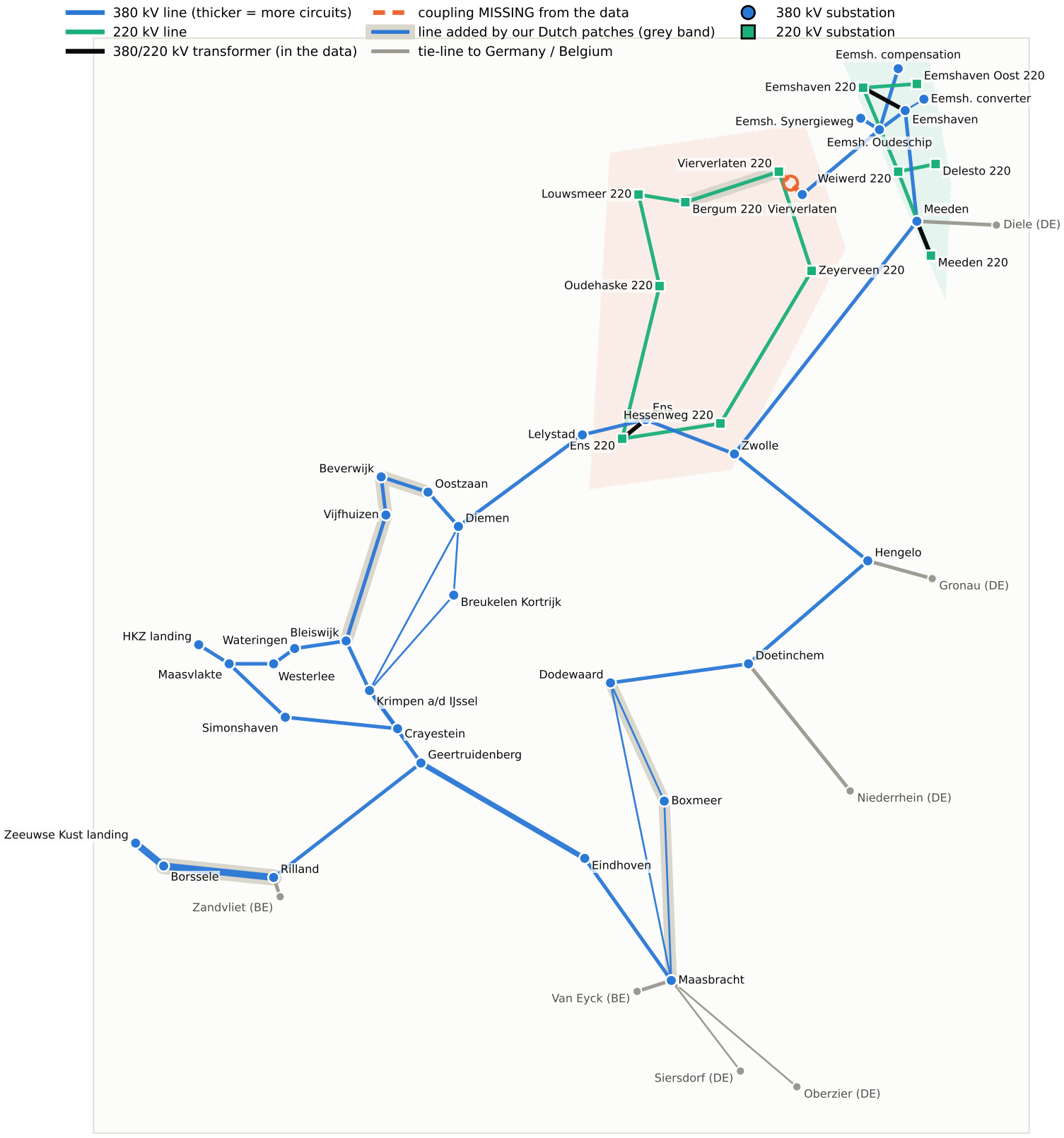
The Dutch transmission grid in the model (v29)

33 substations at 380 kV, 12 at 220 kV, 3 transformer sites; ties to Germany and Belgium. Shaded: the two separate pieces of the 220 kV network.



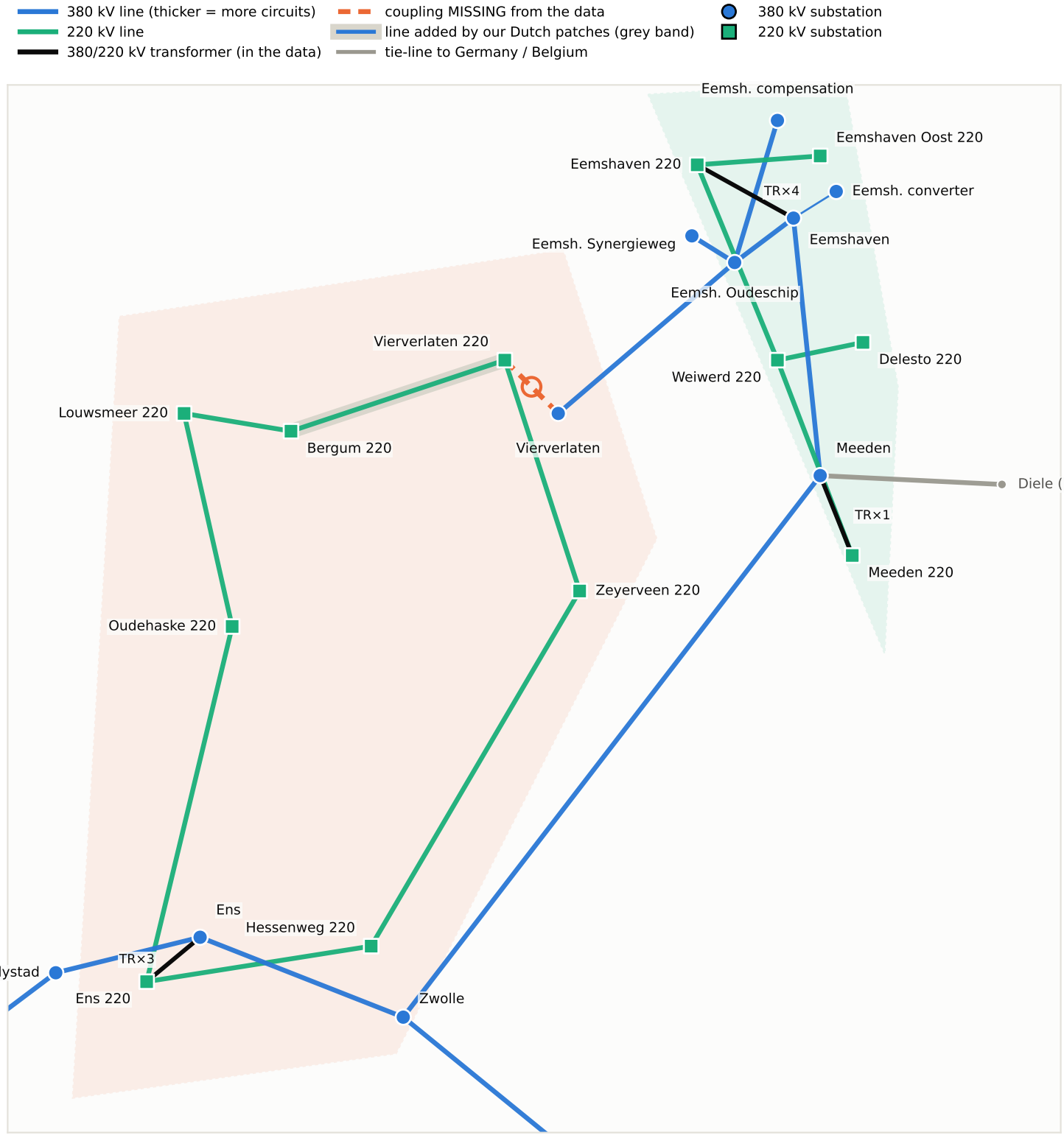
What the map shows

- The 380 kV backbone forms two loops: the Randstad ring (Krimpen– Bleiswijk – Vijfhuizen – Beverwijk – Oostzaan – Diemen – Breukelen – Krimpen, closed in the model by our patch; grey band) and the large outer loop Krimpen– Geertruidenberg – Eindhoven – Maasbracht – Dodewaard – Doetinchem – Hengelo – Zwolle – Ens – Lelystad – Diemen.
- The 220 kV network exists only in the north and is in two pieces. The Groningen piece (aqua shading) is coupled to 380 kV at Eemshaven and Meeden: a real parallel path. The Friesland– Overijssel ring (orange shading) is coupled ONLY at Ens: it hangs on one point, so in a DC power flow it carries no through-flow and the reduction drops it.
- Missing from the data (orange dashed): the Vierverlaten 380/220 kV transformers. With them the Friesland ring would become a 220 kV path parallel to Meeden – Zwolle – Ens. Vierverlaten 380 is today a dead end, connected only to Eemshaven Oudeschip.
- Lines added by our Dutch patches (grey band): Beverwijk– Vijfhuizen – Bleiswijk and Beverwijk – Oostzaan, Bergum – Vierverlaten 220 kV, the Maasbracht – Boxmeer – Dodewaard circuit and Borssele – Rilland circuit P (drawn together with the existing Borssele – Rilland circuits).

Workbook v29 (inputs/derived/jao_core_v29); scripts/map_nl_grid_v2.py, 2026-10-19. Positions approximate (0.01°); co-located buses drawn apart; no coastline.

The north: where 380 kV and 220 kV meet

Eemshaven, Meeden, Ens: transformers in the data (black, with the number of units). Vierverlaten: coupling missing (orange dashed). Zwolle and Hessenweg share a site but are not coupled.



Why the missing couplings matter for the nodal model

- Groningen 220 kV piece (Delesto, Eemshaven Oost, Eemshaven, Meeden, Weiwerd): coupled at Eemshaven (4 transformers) and Meeden (1). Already in the model: the Eemshaven – Meeden branch of the nodal network carries 0.93 of the physical line's reactance because this path runs in parallel.
- Friesland – Overijssel ring (Bergum, Ens, Hessenweg, Louwsmeer, Oudehaske, Vierverlaten, Zeyerveen): coupled only at Ens (3 transformers). It carries no through-flow; all load and generation in it must sit at the Ens node.
- With the Vierverlaten coupling the ring would carry part of the north–south flow, relieving Meeden – Zwolle and Zwolle – Ens. In the reduction it would appear as an equivalent Vierverlaten – Ens branch with no direct line, which the present sparse-network rule DROPS: that rule must keep equivalent branches formed through the Dutch 220 kV network before this can show. 220 kV lines and transformers are not monitored, so their own overloads would not limit transfers.
- Data needed: number of transformer units and reactance at Vierverlaten (TenneT); a variant with assumed values exists (workbooks v29-vvltr1/2). Zwolle 380 kV and Hessenweg 220 kV share a site but are NOT coupled (confirmed by the user, 2026-10-19), so the ring couples to 380 kV at Ens and, once added, Vierverlaten only.

Workbook v29 (inputs/derived/jao_core_v29); scripts/map_nl_grid_v2.py, 2026-10-19. Positions approximate (0.01°); co-located buses drawn apart; no coastline.